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Clinical Outcomes of Knee Osteoarthritis Treated With an Autologous Protein Solution Injection: A 1-Year Pilot Double-Blinded Randomized Controlled Trial

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Abstract

Background: Osteoarthritis (OA) is a debilitating disease resulting in substantial pain and functional limitations. A novel blood derivative has been developed to concentrate both growth factors and antagonists of inflammatory cytokines, with promising preliminary findings in terms of safety profile and clinical improvement.

Purpose: To investigate if one intra-articular injection of autologous protein solution (APS) can reduce pain and improve function in patients affected by knee OA in a multicenter, randomized, double-blind, saline-controlled study.

Study design: Randomized controlled trial; Level of evidence, 2.

Methods: Forty-six patients with unilateral knee OA (Kellgren-Lawrence 2 or 3) were randomized into the APS group (n = 31), which received a single ultrasound-guided injection of APS, and the saline (control) group (n = 15), which received a single saline injection. Patient-reported outcomes and adverse events were collected at 2 weeks and at 1, 3, 6, and 12 months through visual analog scale (VAS), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), Knee injury and Osteoarthritis Outcome Score (KOOS), Short Form-36 (SF-36), Clinical Global Impression of Severity/Change (CGI-S/C), Patient Global Impression of Severity/Change (PGI-S/C), and Outcome Measures in Rheumatology-Osteoarthritis Research Society International (OMERACT-OARSI) responder rate. Imaging evaluation was also performed with radiograph and magnetic resonance imaging (MRI) before and after treatment (12 months and 3 and 12 months, respectively).

Results: The safety profile was positive, with no significant differences in frequency and severity of adverse events between groups. The improvement from baseline to 2 weeks and to 1, 3, and 6 months was similar between treatments. At 12 months, improvement in WOMAC pain score was 65% in the APS group and 41% in the saline group (P = .02). There were no significant differences in VAS pain improvement between groups. At 12 months, APS group showed improved SF-36 Bodily Pain subscale (P = .0085) and Role Emotional Health subscale (P = .0410), as well as CGI-C values (P = .01) compared with saline control. Significant differences between groups were detected in change from baseline to 12 months in bone marrow lesion size as assessed on MRI and osteophytes in the central zone of the lateral femoral condyle, both in favor of the APS group (P = .041 and P = .032, respectively). There were no significant differences between APS and control groups in other measured secondary endpoints.

Conclusion: This study provides evidence to support the safety and clinical improvement at 1-year follow-up of a single intra-articular injection of APS in patients affected by knee OA. Treatment with APS or a saline injection provided significant pain relief over the course of the study with differences becoming apparent at between 6 and 12 months after treatment.

Trial registration: [NCT02138890](https://clinicaltrials.gov/ct2/show/study/NCT02138890) (ClinicalTrials.gov identifier).